UX DESIGN STUDIO · CASE STUDY

Closing the Confidence Gap in Public Transport

A five-phase, human-centred design project for newcomers navigating Dehradun's informal transport network — from field research to a tested 12-screen prototype.

User Research | Synthesis | Concept Development | Interaction Design | UI & Design System | Usability Testing

Designer
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Course
UX Design Studio

Location
Dehradun, Uttarakhand, India

Timeline
March – April 2026

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What's inside this case study

The project ran across five phases. Each phase built directly on the evidence produced by the one before it — research findings became insights, insights became principles, principles became screens, and screens were tested with the people they were designed for.

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01	**Phase 1 — Research & Problem Definition**	9 interviews, secondary research, pain-point prioritisation
02	**Phase 2 — Synthesis: The Confidence Gap**	Affinity mapping, 6 insights, 6 principles, 4 design goals
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04	**Phase 4 — Interaction & Interface Design**	IA, wireframes, 8 hi-fi screens, design system, heuristics
05	**Phase 5 — Usability Testing & Iteration**	5 participants, 7 tasks, SUS 74, 9 prioritised iterations
06	**Reflection & Next Steps**	What worked, what I'd change, Round 2 plan

A note on naming

The product carried a working name of **RideMate** through concept and interface design (Phase 4), and was renamed **TransitFlow** for the tested prototype in Phase 5. The underlying concept — *Confidence Companion* — is unchanged throughout. Both names appear in the original phase documents and are preserved here where they were used.

The one-line version

Newcomers to Dehradun don't abandon public transport because route information is missing. They abandon it because nothing in the system ever tells them they are doing the right thing. This project designs that missing confirmation layer.

00 · PROJECT OVERVIEW

The project at a glance

A design studio project addressing why people who are motivated to use Dehradun's public transport still end up in expensive cabs — and what a navigation product would have to do differently to stop that happening.

The problem

Dehradun's transport network is dominated by informal modes — Vikrams (shared autos), shared minivans, e-rickshaws — that no mainstream navigation app covers. For a newcomer, every journey is a series of unverifiable decisions: is this the right vehicle, is this the right fare, is this bus even coming?

The consequence is measurable: overcharging, hours lost waiting, and a permanent switch to ride-hailing at 3–10× the cost.

The insight that reframed it

Nine interviews showed that users were not short of information — they were short of **confidence**. Every workaround they had invented (asking three strangers, keeping a relative on the phone, switching to a cab) was a substitute for a confirmation signal the system never gave.

That reframe — *a confidence gap, not an information gap* — became the spine of the entire project.

My role & the process

Role

Sole designer across all five phases: research planning and interviewing, synthesis, concept generation, information architecture, wireframing, visual design and design system, prototyping, usability test facilitation, and iteration planning.

Methods used

Semi-structured interviews · secondary/desk research · affinity mapping · journey mapping · impact-vs-frequency prioritisation · How Might We reframing · weighted concept comparison matrix · storyboarding · user flow diagrams · component-based design system · Nielsen heuristic evaluation · task-based usability testing with think-aloud · System Usability Scale (SUS).

Headline numbers

9
users interviewed ages 20–50

7/9
named unreliable route info as their #1 challenge

5/9
had been overcharged — ₹50 routes sold at ₹150

4/6
newcomers abandoned PT for a cab mid-journey

12
screens designed and prototyped

74
average SUS score — above the 68 industry average

74%
task completion across 35 task attempts

9
prioritised design iterations produced

The five phases

Phase	Focus	Key output
1 · Research	Problem definition	9 interviews, ecosystem map, 4 key findings, prioritised pain points, initial problem statement
2 · Synthesis	Reframing	The Confidence Gap model, 5 behavioural clusters, 6 insights, 6 design principles, 4 testable design goals
3 · Ideation	Concept development	3 concepts, storyboards, user flows, weighted matrix — Confidence Companion selected at 152/185
4 · Design	Interaction & interface	IA, 8 annotated wireframes, hi-fi UI, design system, heuristic evaluation, accessibility decisions
5 · Testing	Validation & iteration	12-screen prototype tested with 5 newcomers, SUS 74, 9 issues, 9 documented iterations

01

Research & Problem Definition

Understanding user challenges in urban public transportation — Dehradun, India. Primary interviews, secondary transport data, and the first framing of the problem.

- Research overview & methodology
- Who we interviewed — 9 participants
- Four key findings from primary research
- Secondary research — the Dehradun transport ecosystem
- Pain point prioritisation
- Problem statement & design opportunities

PHASE 1 · RESEARCH OVERVIEW

The problem space

Navigating public transport in Indian cities is extremely difficult for unfamiliar users. The absence of reliable route information, unpredictable timings, and non-transparent fares forces travellers to rely on inconsistent local guidance — or to abandon affordable public transport for expensive ride-hailing.

Primary research

9 structured user interviews across diverse demographics — students, working professionals, homemakers, and daily commuters, ages 20 to 50.

Secondary research

Transport authority data, government reports, and urban mobility studies covering Dehradun's fleet composition, fares, and mode share.

Synthesis

Affinity mapping, journey mapping, and impact-vs-frequency prioritisation to turn raw interview material into design direction.

Who we interviewed

Nine participants were deliberately spread across newcomers and long-term locals, so that "confident" behaviour could be compared directly against "lost" behaviour on the same routes.

Participant	Age	Occupation	Travel context
Gaurav Singh	30	Software Engineer	Noida → Dehradun
Nitu Pandey	48	Homemaker	Dehradun → Delhi
Shalu Pandey	26	Homemaker	Dehradun (originally from Nagina)
Minakshi Sharma	40	Teacher	Dehradun → Himachal
Pramod Pandey	50	Service Professional	Delhi ↔ Dehradun
Roodren Patel	25	Student	Gujarat → Dehradun
Tina Raj	20s	Student	Dehradun, daily commuter
Diksha Gogni	24	B.Tech graduate	Dehradun resident, 15+ years
Himakshi Dhami	24	Student	Dehradun outskirts

PHASE 1 · KEY FINDINGS

What the interviews revealed

Finding 1 — No reliable route information

7/9

users reported this as their number one challenge

> *"I changed autos 3 times and constantly asked people for guidance — everyone said something different."*
>
> — Gaurav, 30

Locals give conflicting directions. There is no single reliable source of route information — and asking more people makes the picture worse, not better.

> *"I couldn't understand anything. Had to get someone on a phone call to guide me through Delhi."*
>
> — Nitu, 48

> *"Conflicting directions from different people only added to my confusion."*
>
> — Shalu, 26

Findings 2 & 3 — Unpredictable timing and fare exploitation

Unpredictable waiting

- **2+ hours** — Roodren waited at the airport for a bus that never came on schedule
- **Shared vehicles** only leave once they are fully packed, so departure is never predictable
- **What people missed:** a train, exams, a bank visit, a full day of lectures

Fare exploitation

- **5 out of 9** users reported being overcharged
- **₹50 → ₹150** for ISBT to GMS Road — a 3× markup on a standard route
- **₹1,500 → ₹2,600** for an airport taxi, under a taxi union monopoly

Finding 4 — The cab-switching fallback

When public transport fails, users pay a steep financial penalty. Four distinct triggers pushed people out of the system:

Time urgency

Exam, meeting, or appointment pressure removes any tolerance for uncertainty.

Route confusion

No idea which vehicle goes where — and no way to check before committing.

No night transport

Services stop after dark, leaving no legitimate option at all.

Last-mile gap

Public transport simply cannot reach the final destination.

Result: users switch to cabs at 3–10× the cost

Gaurav booked a cab mid-journey when route confusion peaked. Pramod paid ₹4,000 for an emergency drop. Shalu switches to cabs for comfort when running late. This is a direct financial penalty for an information gap.

PHASE 1 · SECONDARY RESEARCH

Mapping the Dehradun transport ecosystem

To understand what a newcomer is actually facing, the whole network was mapped — every mode, its colour identity, fare band, capacity, frequency, operating hours, and accessibility level.



Dehradun's complete public transport ecosystem — Vikrams, city buses, electric buses, auto rickshaws and Tata Magic shared minivans, with interchange points, fare bands, frequencies and accessibility ratings.

Two things stood out from this map. First, **the Clock Tower (Ghanta Ghar) functions as the network's single universal interchange** — almost every route touches it. Second, **the dominant mode is also the least documented one**: Vikrams carry the most passengers and appear on no mainstream navigation app.

The numbers behind the problem

22%
public transport usage

78%
private vehicle registrations

17
PT vehicles per 1,000 residents

1300%
vehicle growth, 2005–2025

Mode	Fleet size	Fare range	Role in the network
Vikrams (shared auto)	~3,000+	₹10–25	Dominant everyday mode; colour-coded by route; invisible to navigation apps
City buses	250–300	₹10–40	Uttarakhand Roadways; scheduled but unreliable in practice
E-rickshaws	4,000–5,000+	₹10–30	Last-mile and short hops; entirely informal
Ride-hailing	Thousands	₹80–250	The fallback users are pushed into — 3–10× the cost

Why users abandon public transport

Unreliable schedules

20–60 minute waits, with only two buses on some routes.

Limited coverage

Private buses serve only profitable routes; the outskirts are neglected.

Crowding & discomfort

Small-capacity vehicles are packed well beyond limits.

No door-to-door option

Multiple transfers, compared with direct private travel.

Mode share context: 43% / 35% / 22% split across vehicle registration categories, against just 22% public transport usage — the gap between what the city owns and what the city rides.

PHASE 1 · PRIORITISATION & FRAMING

Pain point prioritisation

All observed pain points were plotted on an impact-versus-frequency matrix, so that design effort could be spent where harm is both common and severe rather than where it is merely visible.

Priority band	Pain points
CRITICAL	No real-time information / unpredictable timings · Overcharging of outsiders · Conflicting route guidance
HIGH	Delays causing real losses · No disruption alerts · Last-mile connectivity gap
MODERATE	Overcrowding · Driver accountability · Cash-only payment and no night service
RARE BUT SEVERE	Vehicle safety neglect · Digital literacy barrier · Airport taxi monopoly

Problem statement (Phase 1)

People travelling in or to Dehradun face significant difficulty navigating the city's public transport system due to the absence of reliable route information, unpredictable vehicle timings, and non-transparent fare structures.

This forces users — particularly visitors, students from outside the city, and less digitally literate individuals — to rely on inconsistent guidance, experience costly delays, fall victim to overcharging, and frequently abandon affordable public transport for expensive ride-hailing alternatives.

Supported by: 9 user interviews · 7/9 cited the information gap as their #1 problem · 5/9 overcharged · all participants wanted a digital solution.

Design opportunities identified

Real-time vehicle tracking

Live bus and Vikram locations with ETAs — addresses the #1 pain point directly.

Multi-modal route planning

An integrated journey planner across buses, Vikrams and e-rickshaws, with transfer guidance.

Transparent fare display

Standardised fare information for any route, pre-empting the exploitation of visitors.

Disruption alerts

Proactive notifications for strikes, blockages and diversions — with alternatives attached, not just warnings.

Accessibility-first design

Simple UI, minimal steps, and vernacular language support for all literacy levels.

Where Phase 1 left the project

This is a solid but generic conclusion — every Indian city produces a report that says "no route info, no timings, no fare transparency." Phase 2 asked a harder question: *what is actually happening in these stories that a competitor's report would miss?*

02

Synthesis — The Confidence Gap

Why users don't fail because they lack information — they fail because they lack confidence in their decisions.

- Surface reading vs. the deeper pattern
- Evidence from research — user voices
- Reframed problem statement
- Affinity map — five behavioural clusters
- Six synthesised insights & the Confidence Gap model
- How Might We questions and five opportunity areas
- Six design principles and four design goals

PHASE 2 · THE REFRAME

What most people see vs. what the data actually says

The same nine interviews support two very different readings. The first is the one every transport report produces. The second is the one that came out of reading the interviews for behaviour rather than for complaints.

Surface level — what most reports say

- No route information available
- No timing data for buses or Vikrams
- Users are confused about routes
- The city is missing a navigation app

Generic. Every city produces the same report. Not a differentiator.

Deeper pattern — what the data is really saying

- Users don't trust the system
- Users depend on other humans to verify their decisions
- Users constantly doubt decisions they have already made
- Cab-switching is a loss of confidence, not a lack of route information

Unique to this research. This is the differentiator.

The hidden insight — The Confidence Gap

The real problem is not missing information. The real problem is missing confidence while making decisions.

This is psychology and behaviour, not cartography — and it is what separates this project from a map.

"Everyone said something different"

= need for reassurance

"I switched to a cab because I wasn't sure"

= loss of confidence

"Didn't know if I'm on the correct route"

= fear of a mistake

PHASE 2 · EVIDENCE

User voices — every quote maps to a confidence failure

Each interview quote was re-coded against the type of confidence failure it represented, rather than against the surface complaint it described.

CONFIRMATION FAILURE

"I changed autos 3 times, constantly asked people — everyone said something different."

Gaurav, 30 — Software Engineer, Noida → Dehradun

Needed reassurance at every step — not a map, a verifier.

HUMAN DEPENDENCY

"I couldn't understand anything. Had to get someone on a call to guide me through."

Nitu, 48 — Homemaker, Dehradun → Delhi

Used a person as a live confidence engine — because no system provided one.

UNCERTAINTY PARALYSIS

"Didn't know if the bus was coming. Waited 2+ hours. Missed my entire day."

Roodren, 25 — Student, Gujarat → Dehradun

Could not make the switch decision — no signal to act on. Paralysis by uncertainty.

CONFIDENCE LOSS → CAB EXIT

"I switched to a cab because I wasn't sure I was on the right path."

Shalu, 26 — Homemaker, from Nagina

The cab wasn't chosen for comfort — it was chosen because doubt became unbearable.

ZERO DISRUPTION FEEDBACK

"I had no idea about the lawyers' strike. I reached the stop and only then found out."

Tina, 20s — Student, daily Dehradun commuter

Even a local expert received zero signal about a route disruption.

REAL-COST CONSEQUENCE

"The bus kept stopping every 100–200 metres. I arrived 2 hours late. The bank was closed."

Pramod, 50 — Service Professional, Delhi ↔ Dehradun

Unpredictability turned into a missed financial transaction. The stakes are real.

PHASE 2 · REFRAMED PROBLEM

From a generic framing to a specific one

Old framing (generic)

"Users cannot navigate public transport in Dehradun due to lack of route information, unpredictable timings, and non-transparent fares."

New framing (unique)

"Users lack confidence while making decisions in public transport due to uncertainty, inconsistent information, and the absence of reliable feedback during the journey."

Full problem statement

Newcomers to Dehradun — visitors, out-of-city students, and recent migrants — are systematically excluded from the city's public transport not because routes don't exist, but because the system provides zero confidence signals during the journey. Users cannot verify they are on the correct vehicle, cannot trust a quoted fare, cannot know if their bus is coming, and receive no feedback when something goes wrong. The result is decision paralysis, dependency on strangers for reassurance, and permanent exit to expensive cabs — even from users who are motivated to use public transport.

6/9
cited the information gap as their #1 problem

5/9
were overcharged

4/6
newcomers cab-switched

0
apps give in-journey feedback

Affinity map — five behavioural clusters

Observations were grouped by *confidence-failure type* rather than by surface problem. Five clusters emerged, and each one later became a design goal.

1 · The Verification Loop

FAILURE: CONFIRMATION

Users ask multiple people not because they want information — they want confirmation that they're doing the right thing. Each answer should resolve doubt. Instead it multiplies it.

"Everyone said something different"

2 · The Paralysis Wait

FAILURE: UNCERTAINTY

No ETA means no decision. Users cannot decide whether to wait or leave because there is zero feedback. They are frozen — until urgency forces the cab.

"Didn't know if the bus will come"

3 · The Fare Ambush

FAILURE: TRUST

Newcomers accept any quoted fare because they have no reference to challenge it. This isn't just financial harm — it's a trust-destruction event that makes them unlikely to use PT again.

"₹50 route charged as ₹150"

4 · The Stress Exit

FAILURE: DECISION COLLAPSE

Switching to a cab is not a preference — it is a surrender. When confidence collapses under time pressure, users exit the system permanently. They don't return.

"I wasn't sure, so I booked a cab"

5 · The Silent Disruption

FAILURE: BLIND JOURNEY

Even a motivated, experienced local received zero warning about a route blockage. Mid-journey disruptions are invisible; users only discover problems after they have already happened.

"Only found out when I reached the stop"

PHASE 2 · INSIGHTS

Six insights, each with a design implication

I1 · The trust gap is a design failure, not a user failure

Users are applying learned scam-detection behaviour correctly. The system has done nothing to distinguish itself as trustworthy. This is not user ignorance — it is design negligence.

→ Every touchpoint must carry a verifiable trust signal.

I2 · Confidence, not information, drives mode choice

Users switch to cabs not when they lack route information, but when their confidence in their current decision collapses. A route they half-know feels worse than a cab they're certain about.

→ Design for confidence at every decision point, not just information provision.

I3 · Humans are the current confidence engine

Every user compensates for the system's lack of feedback by using another human as a live confidence validator — locals, drivers, relatives on phone calls. Design must replicate this function.

→ Replace human dependency with system-level reassurance signals.

I4 · Mid-journey is the highest-risk, lowest-served moment

Most apps solve pre-journey planning. But the confidence collapse happens during the journey — wrong vehicle, unexpected stop, overcharging moment. Zero apps serve this window.

→ Prioritise mid-journey feedback: 'You are on the correct vehicle.'

I5 · Fare opacity is a trust-destruction event

Being overcharged does not just cost money — it permanently damages trust in the entire PT system. Users who experience fare exploitation don't re-try. Each overcharge is a user lost forever.

→ Make fare verification available before boarding, not after.

I6 · Local knowledge proves confidence is achievable

Tina and Diksha navigate confidently at low cost — without an app. Their confidence comes from experience, not from the system. The opportunity is to transfer that feeling, not just the data.

→ The goal is to deliver the feeling of 'I know what I'm doing' — not just a map.

The Confidence Gap model — how failure actually happens

Plotting emotional state against journey stage showed exactly where users are lost — and it is not at the planning stage that every existing app optimises for.

TRIP PLANNING	ARRIVES AT STOP	BOARDS VEHICLE	MID-JOURNEY	DECISION POINT
Optimistic	**Uncertain**	**Doubting**	**Anxious**	**Collapsing**
Route looks plausible; the user commits.	No ETA — 'should I wait?'	No confirmation — 'am I correct?'	No reassurance — 'is this the right direction?'	No fallback — 'what do I do now?'

The Confidence Gap → the cab exit happens here. Between boarding and the decision point, the user has received no signal of any kind. The current outcome is a permanent cab exit; the design opportunity is confidence restoration at each gap; the measurement is a first trip completed without a cab switch.

PHASE 2 · FRAMING THE DESIGN WORK

How Might We — six design questions

Each HMW question is anchored to a specific insight and a specific failure moment, so that no design idea could drift away from the evidence.

HMW 01

Help a newcomer confirm they are on the correct vehicle — in real time, without asking anyone?

↳ I4 — mid-journey is highest risk

HMW 02

Give a user waiting at a stop a reliable signal to decide — wait, switch route, or take a cab — within 10 minutes?

↳ I2 — confidence drives mode choice

HMW 03

Make the correct fare for any Vikram or auto route verifiable before the user boards — ending verbal negotiation?

↳ I5 — fare opacity destroys trust

HMW 04

Replicate what a trusted local does — reassure, guide, confirm — but through a system instead of a person?

↳ I3 — humans are the confidence engine

HMW 05

Restore a newcomer's confidence after a failed or confusing experience — so they attempt PT again instead of exiting permanently?

↳ I6 — local knowledge shows it's achievable

HMW 06

Alert a user to route disruptions before they arrive at a stop — so they can reroute with confidence, not panic?

↳ Tina — the silent disruption pattern

Five design opportunity areas

Not solutions — validated design spaces derived from the Confidence Gap model.

Opportunity area	What it does	How it differs from what exists	Scope
O1 · Decision Confirmation Layer	Tell users at every step: 'Yes, you are on the correct vehicle.' Replicate digitally what a local friend does.	Maps show routes. This *confirms* you're on the right one.	In-journey, real-time
O2 · Human-Like Guidance System	Replicate exactly what locals do — guide, reassure, confirm. 'Ask the conductor' becomes 'the system tells you'.	Replicates the emotional role of a local guide, not just data.	Onboarding + real-time
O3 · Uncertainty Reduction at Stops	A confidence signal while waiting: 'High certainty — bus coming in ~12 min' or 'Low certainty — consider alternatives.'	Gives a confidence level, not just a location.	Pre-boarding, stop-level
O4 · Mid-Journey Support UNIQUE	Solves during the trip: 'You are on the correct path', 'Prepare to change at next stop', 'Route blocked ahead — exit here'.	No existing app does this for Dehradun's informal transit.	During journey — highest leverage
O5 · Smart Fallback & Recovery	When something goes wrong, don't leave users in panic: 'Running late → faster option: auto from here ₹45.'	Existing apps show an error. This shows the next confident step.	Failure states + recovery

PHASE 2 · PRINCIPLES & GOALS

Six design principles

These principles constrain *what* gets designed, not just how it looks. Every screen in Phase 4 is traceable back to at least one of them.

P1 · Reassure before you inform

Users need to know they are right before they need to know where to go next. Confirmation comes before direction. Every screen should answer 'Am I correct?' first.

From I3 — humans are the confidence engine

P2 · Serve the mid-journey moment

Pre-journey planning is solved by Google Maps, imperfectly. The gap is mid-journey — when users are already in motion and doubting themselves.

From I4 — mid-journey is highest risk

P3 · Confidence over completeness

One route a user deeply trusts beats fifty routes they can't verify. Don't map the whole city. Make one journey feel certain — then expand.

From I2 — confidence drives mode choice

P4 · Replace the human, keep the warmth

The local guide is warm, patient and adaptive. The design should feel the same — not a cold info display, but a companion that checks in, confirms, and redirects.

From I3 — humans are the confidence engine

P5 · Make failure a decision, not a dead end

When something goes wrong, don't leave users with an error. Give them the next confident step: 'Bus delayed — here's your fastest alternative right now.'

From the stress-exit pattern

P6 · Earn trust before asking for action

Show the accurate fare. Show the correct vehicle. Show a real ETA. Only after establishing credibility can the system ask users to trust its guidance mid-journey.

From I5 — fare opacity destroys trust

Four design goals — what success looks like

Specific, testable outcomes — one per confidence-failure type. These goals carried through Phases 3, 4 and 5 unchanged, and became the pass/fail criteria for the prototype.

Goal	Target	Definition	Solves
DG1	Confirm vehicle in under 60 seconds	A newcomer boarding a Vikram receives confirmation they are on the correct vehicle in under a minute — without asking a person.	Verification Loop — 7/9 asked multiple people
DG2	Wait decision made within 10 minutes	A user at a stop receives a confidence signal — 'wait' or 'switch' — within 10 minutes of arriving. Not exact GPS; just enough to decide.	Paralysis Wait — Roodren waited 2+ hours
DG3	Fare visible before boarding	A user can see the expected fare for any route before entering a vehicle — eliminating verbal negotiation and pre-empting overcharging.	Fare Ambush — 5/9 overcharged; ₹50 routes at ₹150
DG4	First journey completed without cab fallback	A newcomer completes their first PT trip in Dehradun without being forced into a cab by uncertainty, confusion or panic.	Stress Exit — 4/6 newcomers cab-switched

Positioning — why this is different from Google Maps

Google Maps / existing apps	Confidence-based navigation
Shows a route — leaves the decision entirely to the user	Guides *and* reassures — 'YES, you are on the correct vehicle'
Provides info — gives no reassurance or confirmation	Confirms decisions — 'You are 2 stops away — stay on'
Pre-journey only — zero mid-journey support	Mid-journey support — checks in during the trip, not just before
Doesn't cover Vikrams — informal transit is invisible	Covers Vikrams — the most-used mode, finally legible
Shows error states — no recovery or alternative suggested	Smart fallback — 'Bus delayed → auto option here ₹45'
One-time display — no ongoing confidence signal	Ongoing confidence — a signal at every decision moment

Refined project brief — Phase 2 deliverable

Confidence-Based Navigation for Public Transport. Designing a system that builds confidence in public transport decisions — not just a map. Focus: newcomers to Dehradun — visitors, out-of-city students, recent migrants.

Newcomers are not failing because they lack information — they are failing because the system provides zero confidence signals during the journey. Every human they consult is a workaround for a system that should be doing this job. The design challenge is to build a system that makes users feel *certain* — not just informed.

03

Ideation & Concept Development

Three distinct concepts generated against the Confidence Gap, storyboarded, flow-mapped, and evaluated on a weighted matrix derived from the research.

- Concept sketches — three strategic angles
- Storyboards — Concept A in a real journey
- User flow definition across five journey phases
- Weighted concept comparison matrix
- Selected concept & justification
- Known challenges and planned mitigations

PHASE 3 · CONCEPT SKETCHES

Three concepts, three strategic angles

Each concept addresses the Confidence Gap from a different direction, and each targets a different primary confidence failure — so the comparison would be a genuine strategic choice rather than a styling preference.

Concept A — Confidence Companion

Real-time mid-journey guidance that acts as a trusted local friend.

- In-vehicle confirmation: 'You are on the correct Vikram'
- Proactive stop countdown mid-journey
- Fare lock-in before boarding — ends overcharging
- Smart fallback when confidence collapses
- Disruption alerts before arrival at the stop

Concept B — CrowdSignal

Community-powered real-time transport intelligence for informal transit.

- Live crowd-reported ETA at stops
- Community-verified fare rates for every route
- Crowdsourced strike and disruption alerts
- Route verification via recent user reports

Concept C — StopCoach

Single-screen stop-level confidence assistant for the waiting moment.

- One-tap check-in — no sign-up required
- 3-state signal: WAIT / SWITCH / ALTERNATIVES
- Fare display before boarding at any stop
- Offline-capable for low-connectivity areas

Confidence failures addressed by each concept

Confidence failure	Concept A	Concept B	Concept C
Verification Loop	PRIMARY	PARTIAL	NOT ADDRESSED
Paralysis Wait at stop	STRONG	STRONG	PRIMARY
Fare Ambush	PRE-BOARDING	COMMUNITY	DISPLAY
Silent Disruption	ALERTS	PRIMARY	NOT ADDRESSED
Stress Exit / cab switch	PRIMARY	PARTIAL	DECISION SUPPORT
Human Dependency	PRIMARY	STRONG	PARTIAL

PHASE 3 · STORYBOARD

Concept A in a real journey

Roodren (25, student from Gujarat) navigates Dehradun for the first time using Concept A. Each frame maps to a specific Confidence Gap failure moment identified in Phase 2 research.

#	Frame	What the system shows	What it fixes
1	**Enter destination** Feels: hopeful	Search: Clock Tower → ISBT. 'Have you used this route before?' → No → Companion Mode activates.	The system learns this is a newcomer and will check in at every anxiety point, not just plan a route.
2	**Route + fare locked** First confidence moment	Vikram Route 7 confirmed. Fare ₹15–20 — 'do not pay more'. Board at Clock Tower stop.	**Fare Ambush prevented.** Before Roodren even leaves for the stop, he can screenshot a verified fare.
3	**Arrives at stop** Feels: certain	Status: HIGH CONFIDENCE. Vikram likely arriving in ~8 min, based on recent community reports.	**Paralysis Wait broken.** A wait-vs-switch decision in under 2 minutes, without asking a single person.
4	**Boarding confirmation** Feels: confident	Vikram #7 — route confirmed. 'You are on the correct vehicle.' ISBT via Clock Tower · 2 stops.	**Verification Loop broken.** 7/9 participants asked multiple people at exactly this moment. Roodren doesn't.
5	**Mid-journey check-in** Feels: reassured	On track — 2 stops to ISBT. 'Stay seated. Prepare to exit soon.' Journey 65% complete.	**Mid-journey anxiety dissolved.** The app now performs the role currently played by strangers and phone calls.
6	**Disruption alert** Feels: calm	Route blocked ahead — lawyers' strike, Clock Tower closed. Alternative: exit next stop → auto ₹30.	**Silent Disruption broken.** Alerted before the driver announces it — the opposite of Tina's experience.
7	**Arrival + summary** Feels: proud	Arrived at ISBT. Fare paid ₹18 (within locked range). Cab switches: 0. 'Save route for next time?'	**DG4 achieved** in a single journey — first trip completed without a cab fallback. He will use PT again.

Scenario map — Shalu's fare ambush prevention

Shalu (26) arrives at Clock Tower to take a Vikram to GMS Road. She previously paid ₹150 for this ₹50 route. Today the app shows a confirmed fare of ₹15–25 before boarding → she challenges the driver's ₹150 quote → boards at the correct fare → arrives safely. **Saved ₹130, and rebuilt trust in the PT system.**

PHASE 3 · USER FLOW

Interaction map across five journey phases

Each row records what the user does, how they feel at that moment, and what the Confidence Companion system delivers in response. Emotion is treated as a first-class column, because emotion is what the product is designed to change.

Phase	User action	How the user feels	System touchpoint
Pre-trip	Opens app, enters origin and destination	Hopeful but uncertain — has bad memories of getting lost	Onboarding scan: asks if this is a first time on the route, activates Companion Mode
Pre-trip	Receives route plan with vehicle type and verified fare range	First confidence moment — knows what to expect before leaving	Fare lock: shows ₹15–20 range, vehicle type, stop name
At stop	Checks stop confidence status	Uncertain — should I wait here or leave now?	Stop status: HIGH / MEDIUM / LOW certainty + estimated wait window
At stop	Taps 'Alternatives' if stop status is LOW	Relieved — making an informed decision without guessing	Fallback panel: nearest auto stand, distance, estimated fare
Boarding	Enters or scans the vehicle route number	Doubting — is this actually the right vehicle?	Vehicle confirmation: green tick if correct, red alert if wrong
Mid-journey	Receives proactive stop countdown notification	Anxious — am I going in the right direction?	'2 stops remaining. Stay seated.' — quiet, non-intrusive push
Mid-journey	Disruption detected — route blockage	Panic — what do I do now?	Alert + immediate fallback: 'Exit at next stop, auto to ISBT ₹30'
Arrival	Exits vehicle at the correct stop	Relieved and empowered — I did it without a cab	Journey summary + option to save the route for future trips
Post-trip	Rates the confidence level of the completed journey	Confident — would use public transport again	Feedback feeds route reliability data for future users

Decision architecture — the mid-journey critical branch

When the system detects no movement for more than 3 minutes *or* a route deviation greater than 200 m:

Known traffic delay → 'Slight delay — still on correct route. Relax.' Confidence is restored immediately, with no action required.

Unknown deviation → Alert triggered: 'Possible issue — tap for alternatives.' The user chooses: Wait · Exit · Reroute.

The branch exists because an unexplained silence is what turns doubt into a cab booking. Even a 'nothing is wrong' message is a confidence signal.

PHASE 3 · EVALUATION

Weighted concept comparison matrix

Every criterion is derived from a Phase 2 research insight and weighted by its impact on the Confidence Gap. Scores run 1–5, and each criterion links back to a specific user quote or data point from the nine interviews.

Criterion	Wt	A	B	C	Why this criterion exists	Source
Mid-journey confidence support	5×	**5**	2	2	Highest-risk, lowest-served moment. Zero apps serve mid-journey for Vikrams.	I4 — mid-journey
Vehicle confirmation signal	5×	**5**	3	2	7/9 users asked multiple people when boarding — the Verification Loop failure.	DG1 — 60 sec
Covers informal transit (Vikrams)	5×	**5**	4	3	Vikrams are the dominant mode in Dehradun. Google Maps does not cover them at all.	Positioning
Fare transparency before boarding	4×	**5**	4	4	5/9 overcharged. ₹50 route sold as ₹150. Trust destruction prevents repeat use.	I5 — fare opacity
Stop wait-or-switch decision	4×	4	4	**5**	Roodren waited 2+ hours and missed an entire day. Decision needed in under 10 minutes.	DG2 — 10 min
Low digital literacy accessibility	4×	4	2	**5**	Nitu (48, homemaker) could not use any digital tool independently.	Nitu — I3
Disruption alerts	3×	4	**5**	2	Tina — an experienced local got zero warning about the lawyers' strike until arrival.	Tina — HMW 06
Technical feasibility (MVP)	3×	3	2	**5**	Crowdsourcing needs critical mass. GPS for mid-journey needs infrastructure.	Technical
Emotional warmth / human feel	3×	**5**	3	3	Replace the human, keep the warmth. A local guide is warm, patient, adaptive.	P4 — principle
Total weighted score (max 185)	**36×**	**152**	**100**	**110**	**Concept A leads on the criteria that matter most — mid-journey, vehicle confirmation, and emotional trust.**	

Concept A — Confidence Companion

152 / 185

Leads on mid-journey, vehicle confirmation and warmth.

Concept B — CrowdSignal

100 / 185

Strong on disruption alerts; weak on mid-journey and accessibility.

Concept C — StopCoach

110 / 185

Excellent accessibility and feasibility; limited scope.

PHASE 3 · SELECTED CONCEPT

Concept A — Confidence Companion

A real-time mid-journey guidance system that acts as a trusted local friend — confirming decisions, not just providing information.

Reason 1 — Addresses the primary Phase 2 insight

The Confidence Gap model shows failure happens mid-journey at the Decision Point, not in pre-trip planning. Concept A is the only concept that operates primarily at this stage. Concepts B and C offer supporting features but do not solve the core failure mode.

Reason 2 — Grounded in direct user evidence

Gaurav asked three different people on every leg. Nitu kept family on the phone as a live guide. Shalu switched to a cab when confidence collapsed. Concept A replaces each of these human workarounds with a system that does the same job.

Reason 3 — An unserved design space

Competitive review: Chalo (formal bus only, Tier 1 cities), CityBus Uttarakhand (schedules only), Google Maps (no Vikram coverage). No app provides mid-journey in-vehicle confirmation for Dehradun's informal modes.

Reason 4 — Success is specific and testable

Maps to all four design goals: DG1 (vehicle confirmed <60s), DG2 (wait decision <10 min), DG3 (fare visible before boarding), DG4 (first trip without cab fallback) — all measurable in prototype testing with real newcomers.

All six Phase 2 principles met

P1 Reassure before you inform

P2 Serve the mid-journey moment

P3 Confidence over completeness

P4 Replace the human, keep the warmth

P5 Make failure a decision, not a dead end

P6 Earn trust before asking for action

Design goal mapping

Goal	Target	How Concept A delivers it
DG1	Vehicle confirmed in under 60 seconds	Boarding confirmation screen. Solves the Verification Loop — 7/9 users asked multiple people.
DG2	Wait decision made within 10 minutes	Stop confidence signal HIGH/MEDIUM/LOW plus fallback panel. Solves the Paralysis Wait — Roodren, 2+ hours.
DG3	Fare visible before boarding	Fare lock-in shown in the route plan before the user leaves for the stop. Solves the Fare Ambush — 5/9 overcharged.
DG4	First journey completed without cab fallback	Full Companion Mode: confirmation + countdown + disruption alert + fallback. Solves the Stress Exit — 4/6 newcomers.

Known challenges and planned mitigations

Known challenge	Planned mitigation for Phase 4
GPS accuracy for informal Vikrams — no formal tracking exists	Manual route-number entry as the GPS fallback — no tracking infrastructure required
Data infrastructure needed for real-time stop signals	A community-seeding hybrid (borrowing the Concept B element) to seed stop data
Connectivity gaps in the Dehradun outskirts (Himakshi's route)	Offline-first architecture with cached route and fare data

04

Interaction & Interface Design

Turning the selected concept into a working interface — information architecture, wireframes, interaction states, visual language, and a component-based design system.

- Information architecture & navigation model
- Wireframes — all eight screens, annotated
- Interaction design — states, micro-interactions, transitions
- High-fidelity UI and visual language rationale
- Design system — colour, type, components
- Usability heuristics & accessibility decisions
- Prototype flows and design goal evaluation

PHASE 4 · INFORMATION ARCHITECTURE

App structure and navigation model

Five primary screens, four contextual overlays, and one emergency mode. The structure is deliberately shallow — under stress, depth is a failure mode.

Primary screen	Contains	Why it exists
Home	Confidence cards · quick actions · saved routes	Shows journey confidence state, not just a search box
Route Finder	Step-by-step plan · stop signal · boarding guide · mid-journey · arrival summary	Carries the user through all five journey phases in one thread
Fare Checker	Route fare table · overcharge report · driver rating	Fare evidence available at any moment, including mid-negotiation
I'm Lost	Location detect · safe fare guide · helpline CTA	Emergency mode reachable from every screen in the app
Profile	Saved routes, college selection, Companion Mode settings	Personalisation that pre-loads the routes a student actually uses

Navigation model — five journey phases

Each phase has its own screen, its own emotional state, and its own design goal. The interface changes character as the user's anxiety changes.

PRE-TRIP

Home + Route Finder

Feels: hopeful but uncertain

Goal: lock fare + plan route → DG3

AT STOP

Stop Signal

Feels: uncertain — wait or leave?

Goal: wait/switch decision <10 min → DG2

BOARDING

Boarding Guide

Feels: doubting — right vehicle?

Goal: confirm vehicle <60s → DG1

MID-JOURNEY

Journey Tracker

Feels: anxious — right direction?

Goal: continuous reassurance → DG4

ARRIVAL

Summary Screen

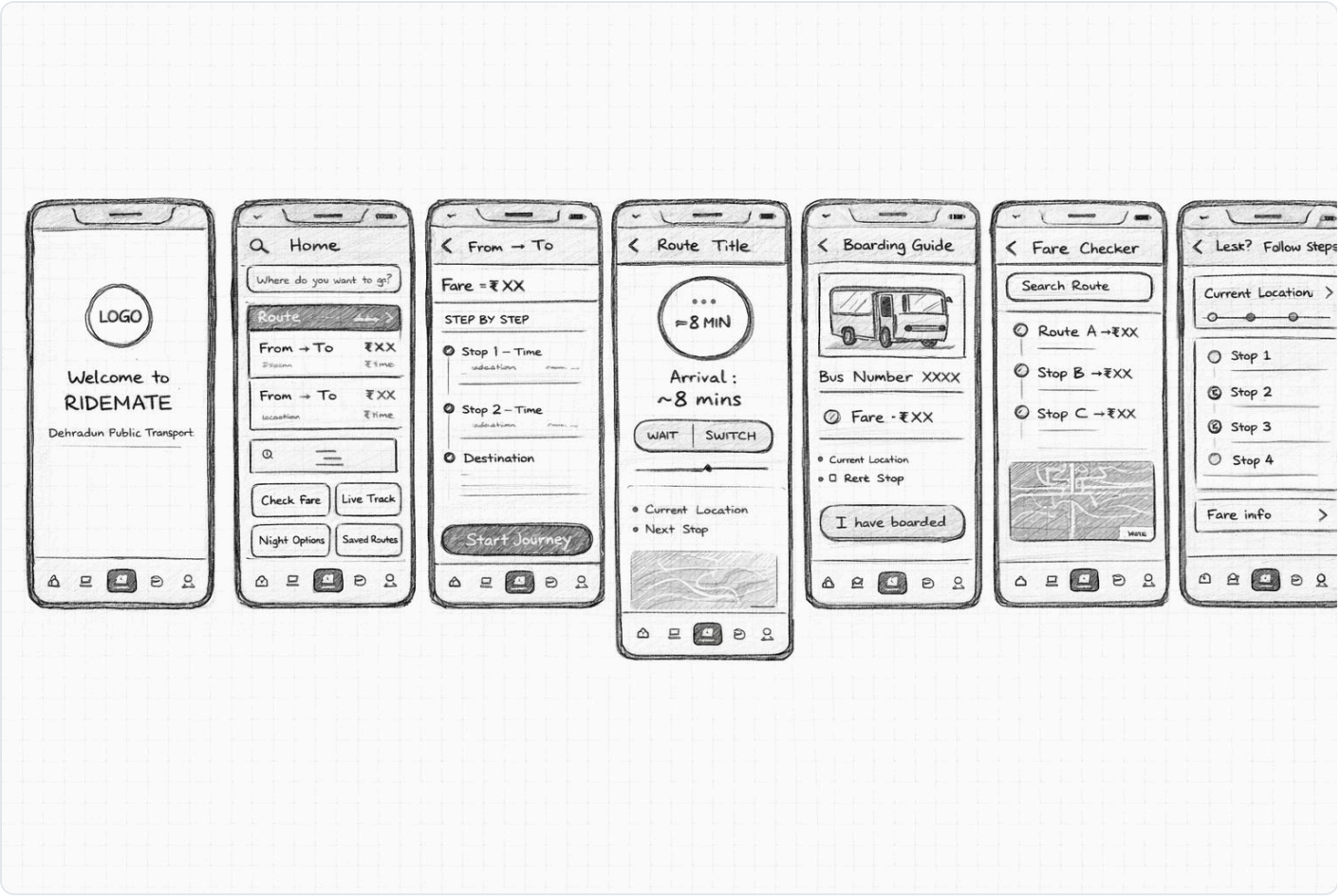
Feels: relieved — I did it

Goal: build confidence for the next trip

PHASE 4 · WIREFRAMES

Low-fidelity wireframes — all screens

Hand-drawn wireframes established the skeleton of every screen before any visual decision was made: welcome, home, route entry, route detail with wait/switch, boarding guide, fare checker, and the lost-recovery flow.



Annotated low-fidelity wireframes — the persistent bottom navigation, the search bar, and the wait/switch decision pair were all fixed at this stage.

What was decided at low fidelity

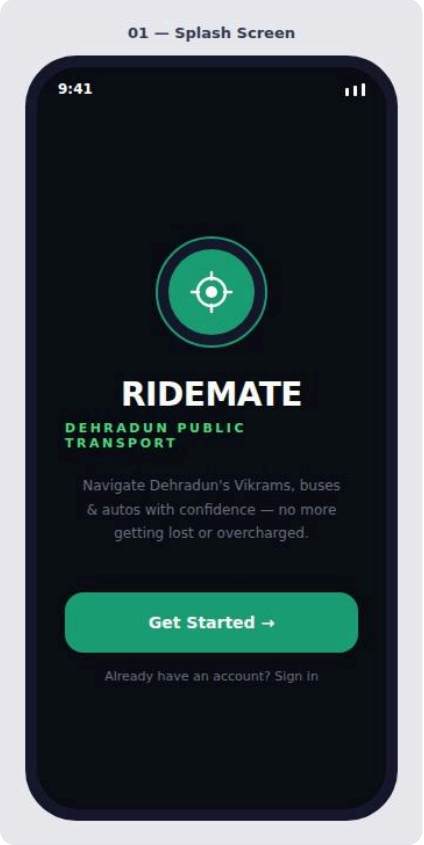
- A persistent five-item bottom navigation, so 'I'm Lost' is never more than one tap away
- Fare shown as a first-class element on every route screen, not inside a detail view
- A binary WAIT / SWITCH pair at the stop — one decision per screen
- 'I have boarded' as an explicit, student-initiated action rather than automatic detection

What was deliberately left out

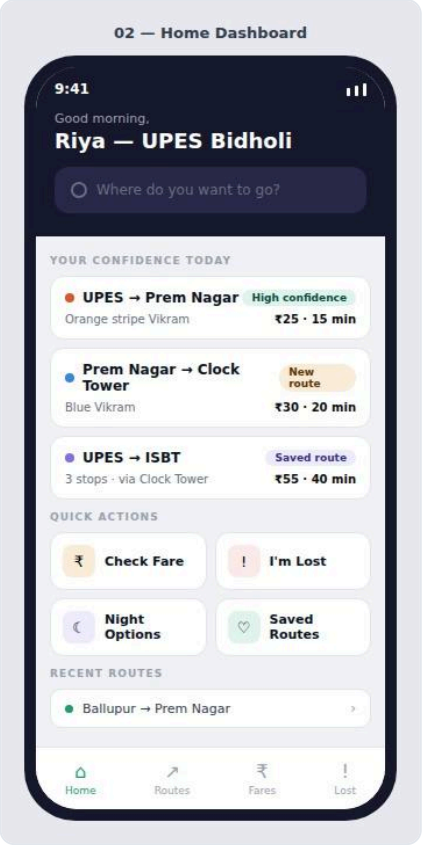
- A map-first home screen — maps answer 'where', but users needed 'am I right?'
- Account creation before first use — friction at the exact moment trust is lowest
- Dense timetable views — completeness at the cost of confidence (P3)

PHASE 4 · HIGH-FIDELITY SCREENS

Screens 1 & 2 — Onboarding and Home Dashboard



Screen 1 — Onboarding



Screen 2 — Home Dashboard

Design decisions — Onboarding

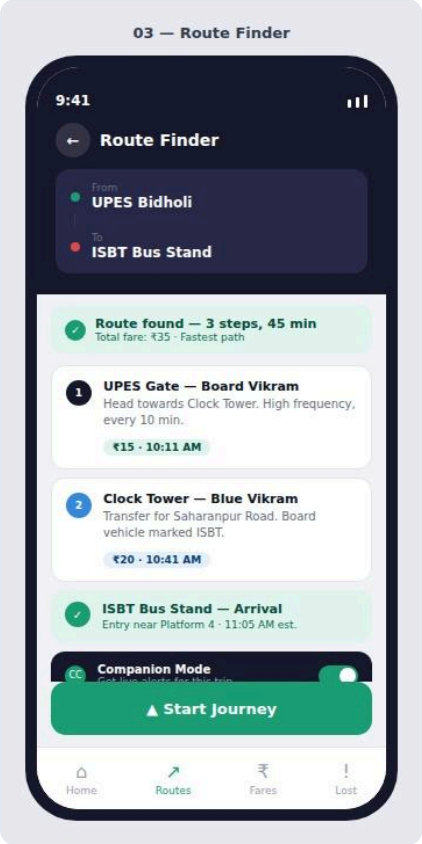
- **College-specific onboarding** — routes pre-load based on the selected institution
- **Four college options** customise default routes and stop locations
- **First-use flow activates Companion Mode** — a higher guidance level for newcomers

Design decisions — Home

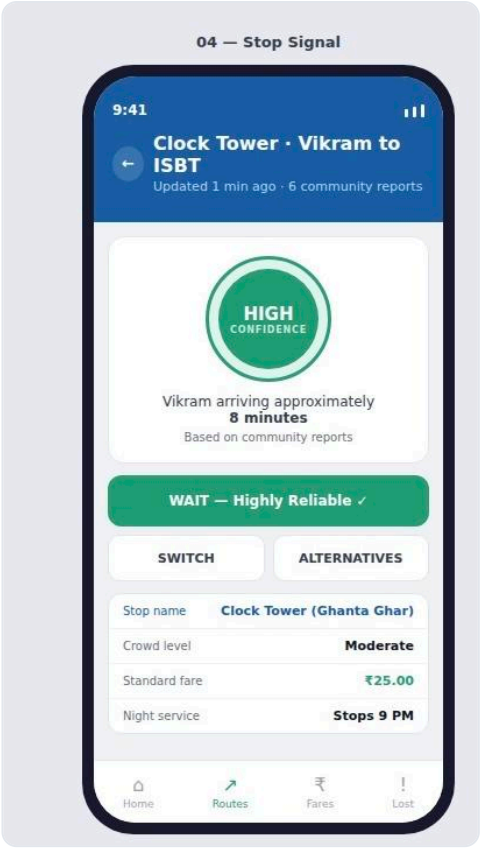
- **Personalised greeting** reduces anxiety — the student feels the system knows them
- **Confidence indicator on every route card** — not just information, a state
- **'I'm Lost' as a primary action** — front and centre, not hidden in settings
- **Persistent search bar** across all screens — always accessible

PHASE 4 · HIGH-FIDELITY SCREENS

Screens 3 & 4 — Route Finder and Stop Confidence Signal



Screen 3 — Route Finder



Screen 4 — Stop Signal

Design decisions — Route Finder

- **Landmark-based input** — 58.8% of users navigate by landmarks, not road names
- **Fare locked before leaving** — DG3 is solved before the journey even starts
- **Step cards carry the exact words to say to the driver**, in plain conversational Hindi
- **Savings displayed** — '₹165 saved vs Ola' addresses budget anxiety directly

Design decisions — Stop Signal

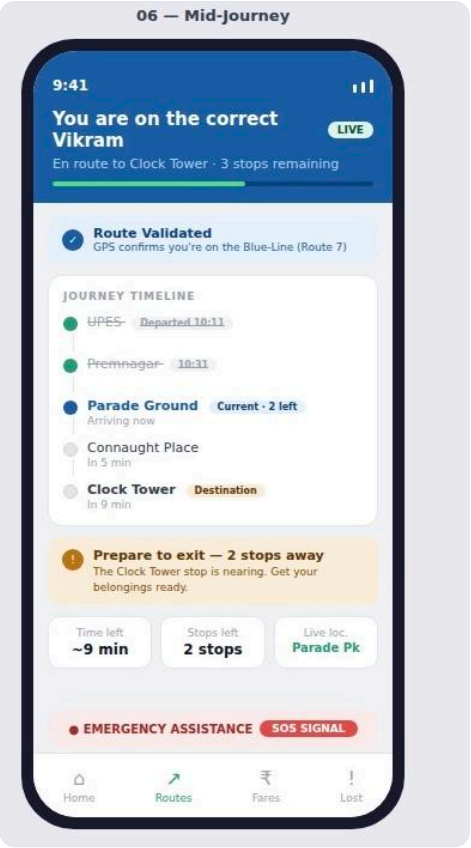
- **Three-state signal** — HIGH / MEDIUM / LOW. Not GPS precision, honest status
- **Wait/switch decision given in under 10 minutes** — DG2 met directly
- **Alternatives always shown** — never a dead end (principle P5)
- Stop metadata: crowd level, standard fare, night-service cut-off

PHASE 4 · HIGH-FIDELITY SCREENS

Screens 5 & 6 — Boarding Guide and Mid-Journey Tracker



Screen 5 — Boarding Guide



Screen 6 — Mid-Journey Tracker

Design decisions — Boarding Guide

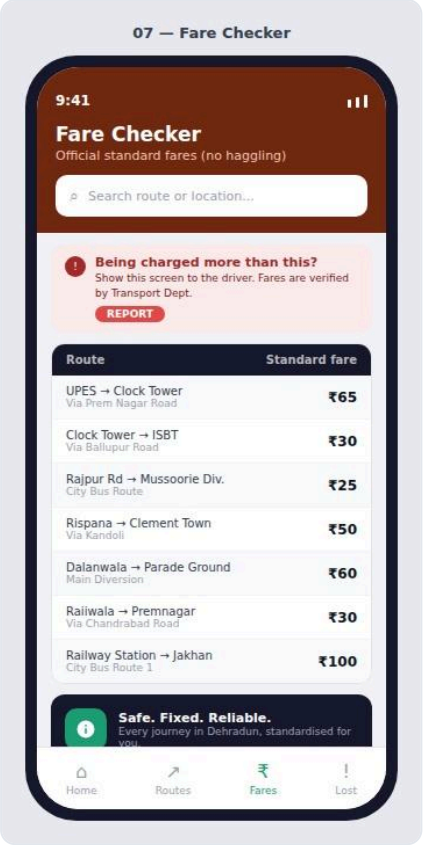
- **Colour stripe is the dominant visual** — match phone to vehicle colour instantly
- **Hindi phrase card** gives the exact words to say, removing social anxiety
- **Fare displayed large** — 'do not pay more than this amount' (DG3)
- **'I have boarded' CTA** activates tracking — student-initiated, not automatic

Design decisions — Mid-Journey

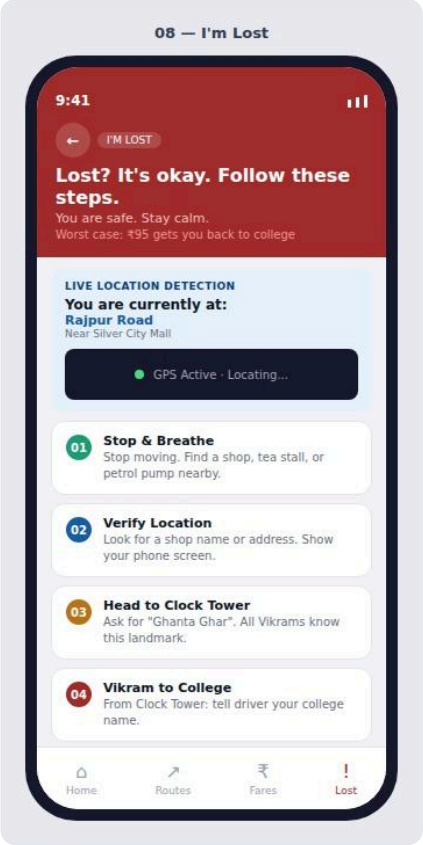
- **'You are on the correct Vikram'** — the single most important message in the app
- **Stop progress tracker** shows exactly where the user is in the journey
- **Proactive alert two stops before the destination** — no panic at the last second
- Live counters: time left, stops left, current location

PHASE 4 · HIGH-FIDELITY SCREENS

Screens 7 & 8 — Fare Checker and I'm Lost (Panic Mode)



Screen 7 — Fare Checker



Screen 8 — I'm Lost (Panic Mode)

Design decisions — Fare Checker

- **The fare table is evidence** — designed to be shown to a driver quoting above standard
- **Report button** feeds community data on exploitative drivers
- **Overcharge banner** positions the phone as the student's protection tool
- Standard fares listed per route (e.g. UPES → Clock Tower ₹65, Clock Tower → ISBT ₹30)

Design decisions — I'm Lost

- **Reassurance first** — 'It's okay. You are safe.' before any instruction
- **Clock Tower as universal reset point** — simplifies any recovery to one step
- **Worst-case cost shown** — a ₹95 cap removes financial panic entirely
- Exactly four recovery steps: stop & breathe · verify location · head to Clock Tower · Vikram to college

A working high-fidelity prototype of these screens was built and shared via QR code for walkthroughs — the version later used for the Phase 5 usability sessions on participants' own phones.

PHASE 4 · INTERACTION DESIGN

Component states

Every interactive element has defined states, so behaviour is predictable and therefore trustworthy. Two components carry most of the product's meaning.

Confidence badge — three variants

HIGH CONFIDENCE

Route is familiar or community-verified.

NEW ROUTE

First time on this route for this user.

UNKNOWN

No data available yet.

Stop confidence signal — three states

HIGH

Wait — Vikram arriving in under 12 minutes.

Action: Wait button active

MEDIUM

Uncertain — the vehicle may be delayed.

Action: Switch shown

LOW

Don't wait — take alternatives.

Action: Alternatives first

Micro-interactions & transitions

Timing and feel were specified alongside behaviour, because in this product the emotional register of a transition is part of the function.

Trigger	Response	Timing	Feels like
Taps 'I have boarded'	Green confirmation banner slides in from the top. Header confidence indicator turns green. Route tracker activates.	300ms ease-in	Achievement — the user feels confirmed
Vehicle confirmed (correct)	Full-screen green flash, then settles to a confirmation card. Haptic feedback pulse.	500ms flash + 200ms settle	Relief — an immediate trust signal
Vehicle confirmed (wrong)	Red alert banner slides in. Gentle haptic error buzz. Clear CTA: 'Exit now — alternatives below.'	400ms slide	Calm correction — not panic
Stop signal updates	Confidence circle transitions smoothly between HIGH/MEDIUM/LOW. Colour fades, never snaps.	800ms colour transition	The system feels alive and aware
Taps 'I'm lost'	Screen slides up as a modal overlay. Reassurance text appears first, then steps fade in sequentially.	200ms overlay + 100ms per step	Calm and measured — not alarming
Arrival detected	Confetti-light animation. Journey summary card slides up. 'Save route?' prompt appears.	600ms celebration	Pride and achievement

PHASE 4 · VISUAL LANGUAGE

Every visual decision derives from a research insight

Element	Rationale	Principle
Navy primary #1A1A2E	Authority and trustworthiness — the opposite of the unreliable informal system students currently experience. They must trust the interface immediately.	P6 — Earn trust before asking for action
Teal green #1D9E75	Confidence and confirmation — used exclusively for positive states: route confirmed, vehicle correct, arrival safe. Builds a Pavlovian trust association.	P1 — Reassure before you inform
Journey blue #185FA5	Journey and progress — the mid-journey screen only. Calming and directional; differentiates the in-motion state from planning or waiting.	P2 — Serve the mid-journey moment
Alert amber #BA7517	Alert without alarm — uncertainty and warnings. Not red (panic), not green (safe). An honest middle ground matching the MEDIUM stop signal.	P5 — Make failure a decision
Emergency red #A32D2D	Emergency only — the I'm Lost screen and the wrong-vehicle alert. Reserved deliberately, so when users see it they know it matters.	P4 — Replace the human, keep the warmth
Inter typeface	Maximum legibility at small sizes — critical for an app used in motion, under stress, in variable lighting, on a moving Vikram.	P3 — Confidence over completeness

Design system — colour palette

Navy primary
#1A1A2E
Headers, navigation, CTA

Confidence green
#1D9E75
Success, confirmation

Journey blue
#185FA5
Mid-journey, information

Alert amber
#BA7517
Warnings, medium confidence

Emergency red
#A32D2D
Panic mode, wrong vehicle

Screen background
#F8F8F8

Surface white
#FFFFFF — cards, panels

Secondary text
#64748B — labels, hints

Border grey
#E5E7EB — dividers

Route identity system — the Vikram colours

Dehradun's Vikrams are already colour-coded in the real world. The product borrows that existing knowledge instead of inventing route numbers — and pairs every colour with a shape, so colour is never the only carrier of meaning.

Colour	Shape	Destination
Orange	Circle	Clock Tower
Blue	Triangle	ISBT
Green	Square	Premnagar
Purple	Diamond	Sahastradhara
Amber	Star	GMS Road

Typography — Inter

Screen title 18px semibold · Card heading 15px medium · Body text 13px regular · Caption label 10px medium uppercase. Noto Sans Devanagari is used at 16px wherever Hindi script appears.

PHASE 4 · DESIGN SYSTEM

Core components

Built as a component library — atoms → molecules → organisms → screens — so that a screen added later inherits the confidence language automatically.

Component	Level	Specification
Confidence badge	Atom	20px height · 8px horizontal padding · 20px radius · semibold 9px · three variants (High / New / Unknown)
Colour stripe	Atom	Fill width · 6px height · 3px radius · one per route colour — the core visual identity element
Primary button	Atom	Fill width · 48px height · 12px radius · three colour variants (navy / green / red) · medium 13px white text
Route card	Molecule	Fill width · auto height ~76px · 0.5px border · 12px radius · 12px padding · contains badge, dot, line, fare
Step card	Molecule	Fill width · auto height ~80px · horizontal layout: 22px navy number circle + content column
Confirmation banner	Molecule	Fill width · auto ~44px · green tint background · 10px padding · tick circle + text column
Stop progress	Molecule	Vertical stack · 10px circles · three states (done / current / future) · 1px connector lines between stops
Hindi phrase card	Molecule	Fill width · white background · 12px padding · Noto Devanagari 16px for Hindi · English transliteration below

Usability evaluation — Nielsen's ten heuristics

H1 Visibility of system status

The stop signal shows HIGH/MEDIUM/LOW at all times. The mid-journey screen shows 'You are on the correct Vikram' plus a stop countdown. The user always knows where they are.

H2 Match between system and real world

Language mirrors how students actually speak — 'Clock Tower chaloge?' not 'Route 14B'. Landmark names match local knowledge; colour stripes match real vehicle markings.

H3 **User control & freedom**

Every screen has a clear back action. Companion Mode can be turned off. Routes can be searched fresh at any point. 'I'm Lost' is accessible everywhere.

H4 **Consistency & standards**

The navy header across all screens provides consistent orientation. Teal always means positive; red always means emergency.

H5 **Error prevention**

Fare locked before boarding prevents overcharge disputes. An 'Are you sure?' confirmation guards panic mode. Vehicle confirmation happens before the journey starts.

H6 **Recognition rather than recall**

Colour stripes remove the need to remember route numbers. Hindi phrase cards remove the need to know what to say. Stop progress shows past stops, not just the next one.

H7 **Flexibility & efficiency of use**

Saved routes for frequent journeys. Companion Mode auto-deactivates after 3 trips on the same route. Power users can skip onboarding steps on familiar routes.

H8 **Aesthetic & minimalist design**

Each screen shows only what that journey phase needs. No decorative elements. The Hindi phrase is displayed as primary content, not buried in settings.

H9 **Help users recognise and recover from errors**

The wrong-vehicle alert gives an immediate next step, not just an error message. Disruption alerts include alternatives before the user reaches the blocked stop.

H10 **Help & documentation**

The 'I'm Lost' screen *is* the built-in help — no documentation needed. Step-by-step route guidance removes the need for external help.

PHASE 4 · ACCESSIBILITY

Inclusive design decisions

Designed for all literacy levels, all phone types, and all stress levels — because the users least served by the current system are exactly the ones with the least digital confidence.

Visual accessibility

- → Minimum contrast ratio 4.5:1 on all text-background combinations
- → The colour identity system uses shape in addition to colour — never colour alone
- → Minimum 13px body text — readable at arm's length, in motion
- → Bold weight reserved for critical information: vehicle, fare, stop name

Language accessibility

- → Hindi as the primary language for all spoken interaction phrases
- → Noto Sans Devanagari for accurate Hindi script rendering
- → English transliteration below every Hindi phrase — bilingual support
- → Landmark names in local usage — 'Ghanta Ghar', not 'Clock Tower intersection'

Cognitive accessibility

- → One decision per screen — never multiple options competing for attention
- → Progress always visible — the user always knows how far through the journey they are
- → Simple language — 'Stay seated', not 'Remain in vehicle until destination'
- → Worst-case scenario displayed (₹95) — removes the cognitive load of 'what if'

Low-tech accessibility

- → Offline-first architecture — cached routes and fares work without data
- → Manual route entry as a GPS fallback — no tracking infrastructure required
- → Designed for mid-range Android, the most common student device
- → Minimum tap target 44px — usable inside a moving Vikram

Prototype flow — the complete user journey

Roodren (25, from Gujarat) — first journey from UPES to ISBT using Confidence Companion.

Phase	Screen	What happens	Feels
Pre-trip	Onboarding	Selects UPES; activates Companion Mode	Hopeful
Pre-trip	Home	Sees UPES routes; taps 'Route Finder'	Curious
Pre-trip	Route Finder	Enters ISBT; reads the two-step plan with locked fare	Informed
At stop	Stop Signal	Arrives at the stop; sees HIGH confidence	Certain
Boarding	Boarding Guide	Matches the orange stripe; boards the correct Vikram	Confident
Mid-journey	Mid-Journey Tracker	Sees confirmation; follows the stop tracker	Reassured
Mid-journey	Disruption Alert	Strike alert received; chooses the alternative	Calm
Arrival	Arrival Summary	Arrives at ISBT; saves the route	Proud

PHASE 4 · SECONDARY & TERTIARY FLOWS

Two more journeys the product has to survive

Secondary flow — fare ambush prevention (Shalu's scenario)

Shalu (26) previously paid ₹150 for a ₹50 route. Today she uses the Fare Checker before boarding.

#	Screen	Action	Result
1	Home screen	Opens the app at the Clock Tower stop	Taps 'Fare Checker'
2	Fare Checker	Searches the GMS Road route	App shows a ₹15–25 range
3	Fare Checker shown	The auto driver quotes ₹150	Shalu shows the fare screen to the driver
4	Boarding Guide	Driver adjusts to ₹20	Shalu boards at the correct fare
5	Report screen	Reports the overcharge attempt	Data feeds the community drivers list
6	Arrival Summary	Arrives at GMS Road safely	**Saved ₹130. Trust in the PT system restored.**

Tertiary flow — panic mode recovery (Gaurav's scenario)

Gaurav is on a Vikram and realises mid-journey that he is going the wrong way. He taps 'I'm Lost'.

The four steps

- **1 · Mid-journey doubt** — taps 'I'm Lost' from any screen
- **2 · Panic screen activates** — 'It's okay' comes before anything else
- **3 · Location detected** — the app finds the current area
- **4 · Recovery plan shown** — Clock Tower as reset point, max fare ₹95

Design decisions in this flow

- **Reassurance before information** — panic reduces first, then the brain can process steps
- **Clock Tower as universal anchor** — one reset point for the entire city; simplicity under stress
- **Worst-case cost displayed** — knowing the maximum removes financial anxiety
- **Exactly four steps** — short, sequential, achievable; modelled on emergency protocols
- **Always accessible** — present in every screen's bottom navigation, no matter how deep in the app

PHASE 4 · EVALUATION

Design goal verification

Each Phase 2 design goal, checked against the Phase 4 interaction and interface decisions before any user testing began.

Goal	Target & what it solves	How the interface delivers it	Status
DG1	Vehicle confirmed in under 60 seconds Verification Loop — 7/9 asked multiple people	Boarding Guide (Screen 5). Colour-stripe visual match takes under 5 seconds. The Hindi phrase card eliminates verbal negotiation. 'I have boarded' activates confirmation.	MET
DG2	Wait decision within 10 minutes Paralysis Wait — Roodren waited 2+ hours	Stop Signal (Screen 4). HIGH/MEDIUM/LOW displayed immediately on arrival. Wait, Switch and Alternatives are one tap away. No calculation or guessing required.	MET
DG3	Fare visible before boarding Fare Ambush — 5/9 overcharged	Route Finder (Screen 3) locks the fare range into the route plan before the user leaves for the stop. Fare Checker (Screen 7) shows the full table at any point.	MET
DG4	First journey without cab fallback Stress Exit — 4/6 newcomers cab-switched	Full Companion Mode: fare lock (pre-trip) + stop signal (at stop) + colour boarding + stop tracker (mid-journey) + disruption alerts = zero confidence-collapse points.	MET

Known challenges & design mitigations

Challenge	Mitigation built into the Phase 4 screens
GPS accuracy for informal Vikrams — no formal tracking exists for Dehradun's Vikrams and shared autos, so real-time location data is unreliable.	Manual route-number entry as the primary input: the user enters the number seen on the vehicle. GPS is secondary confirmation only. The fallback works with no GPS at all.
Data infrastructure — the stop confidence signal needs real-time updates, and there is a cold-start problem for new routes and stops.	A hybrid seed model: initial data hand-collected for the 20 highest-traffic student routes, with community reports layered on top. A 'data from X minutes ago' timestamp shows signal freshness.
Connectivity gaps in the outskirts — Himakshi's route (Premnagar, UPES Bidholi) has low data coverage in certain stretches.	Offline-first architecture. Route plans, fare data and Hindi phrases are cached locally at install. Core screens work fully offline; only the stop signal needs connectivity.
Low digital literacy users — Nitu (48, homemaker) could not use any digital tool independently, and Hindi literacy varies among hill-district users.	Companion Mode offers voice output. Hindi script appears on all key screens. Minimal text — colour and shape carry primary meaning. One-thumb operation throughout, no complex gestures.

Phase 4 summary

8 screens

Onboarding, Home, Route Finder, Stop Signal, Boarding Guide, Mid-Journey, Fare Checker, Panic Mode

4 design goals

All four verified against specific screen decisions

10 heuristics

Every Nielsen heuristic evaluated and addressed

6 principles

P1–P6 each mapped to specific screen decisions

Phase 5 ready

Three-round testing protocol defined with real newcomers

The testing protocol planned at the end of Phase 4

Round	Method	Participants	Measure
1 · Concept walkthrough	Paper prototype walkthrough, screen-by-screen narration. Does each screen communicate its purpose within 10 seconds?	5 students — first semester, non-Uttarakhand	Comprehension rate per screen
2 · Task-based testing	Figma clickable prototype. Three tasks: plan UPES→ISBT, identify the correct Vikram at Clock Tower, recover from the I'm Lost state.	8 students — mix of semesters and origins	Task completion rate · time on task
3 · Field testing	Real journey with the prototype. Accompany a student on a first unfamiliar Vikram route; observe confidence signals in the real environment; capture think-aloud.	3 students — genuine newcomers, first unfamiliar route	Cab switch rate · self-reported confidence 1–5

05

Usability Testing & Iteration

The 12-screen TransitFlow prototype, tested with five out-of-state students living in Dehradun — task-based testing, think-aloud protocol, and the System Usability Scale.

- The prototype — 12 screens, including uncertainty handling
- Testing plan, participants, tasks and methodology
- Real user feedback, participant by participant
- Data collection — completion matrix and SUS scores
- Analysis — affinity themes and priority matrix
- Nine documented design iterations
- Conclusion and Round 2 recommendations

PHASE 5 · EXECUTIVE SUMMARY

Testing the prototype with real newcomers

This phase evaluates the 12-screen TransitFlow prototype through usability testing with five newcomer students. The prototype is notably mature in handling uncertainty — with dedicated screens for Signal Lost, Route Deviation, Critical Mismatch, Offline Mode and Feeling Unsure — directly addressing earlier feedback that the flows were over-controlled and too ideal.

Testing focused on three questions: do the data-uncertainty screens work as intended, are the confidence indicators (98%, Medium, Low) interpreted correctly, and where do students still struggle?

5	74	12	9
real participants — all out-of-state	average SUS score (industry avg: 68)	screens tested end to end	issues identified and prioritised

Project	TransitFlow — Dehradun Public Transport Navigation
Prototype	12 screens covering Home, Journey, Verification, Recovery and Arrival
Method	Task-based testing + think-aloud protocol + post-test interview + SUS questionnaire
Participants	5 out-of-state college students living in Dehradun (1–8 months' residence)
Date	April 2026

The prototype — 12 screens tested

#	Screen	Purpose	What it tests
01	**Home**	Route entry, eco/time preview, Companion Mode start	Pre-trip
02	**Live Journey**	Journey overview with vehicle, fare, and 98% confidence	DG1, DG3
03	**Clock Tower Stop**	Medium-confidence stop signal with fluctuating data	DG2 + uncertainty
04	**Route Verification**	Low-confidence route — ask driver, check passengers	Data gap handling
05	**Boarding Confirmation**	Medium-confidence vehicle entry by route number	DG1 + verification
06	**Critical Mismatch**	Wrong route detected — recheck or board anyway	Error recovery
07	**Live Tracking**	Mid-journey on-track view with map and stops	Mid-journey
08	**Signal Lost**	GPS lost for 3 minutes — wait/ask/exit decision	Data gap handling
09	**Route Deviation**	System detects the vehicle off course — auto fallback	Recovery
10	**Offline Mode**	No internet — cached features only	Connectivity gap
11	**Feeling Unsure**	Hindi phrase + emergency 112 + driver script	Confidence support
12	**Arrival**	Journey complete — fare paid, save route, stats	Post-trip

The key strength of this prototype

Unlike earlier iterations, this prototype explicitly designs for uncertainty. Signal Lost (8), Route Deviation (9), Offline Mode (10) and Feeling Unsure (11) all handle real-world data gaps — directly addressing the earlier critique that the flows were 'too ideal and linear'.

PHASE 5 · TESTING PLAN

Objectives, participants, tasks and method

Testing objectives

→ Whether the 98% confidence indicator on Live Journey (Screen 2) is correctly interpreted as 'reliable'

→ Whether students handle Medium confidence on the stop signal (Screen 3) by waiting or switching appropriately

→ Whether Critical Mismatch (Screen 6) actually prevents users from boarding the wrong route

→ Whether the Signal Lost recovery flow (Screen 8) keeps users calm and decisive

→ Whether the Hindi phrases on Feeling Unsure (Screen 11) are understandable and useful

→ Whether Offline Mode (Screen 10) is trusted as a fallback rather than triggering panic

Participants

ID	Name	Age	College	Home state	Months in Dehradun
P1	**Roodren**	20	UPES	Gujarat	3
P2	**Aarav**	19	Graphic Era	Bihar	2
P3	**Meera**	21	DIT	Tamil Nadu	5
P4	**Karan**	22	UPES	Punjab	8
P5	**Priya**	20	Graphic Era	Madhya Pradesh	1

Test tasks

#	Task description	Screens involved
T1	Plan a route from Clock Tower to ISBT and start Companion Mode	1, 2
T2	At the Clock Tower stop, the signal is fluctuating — decide what to do	3
T3	Verify route 7 before boarding — the system shows Medium confidence	5
T4	Handle a Critical Mismatch alert — the wrong route has been entered	6
T5	Mid-journey, the GPS signal is lost for 3 minutes — recover	8, 9
T6	The internet connection drops — continue using offline features	10
T7	Feeling unsure — use the Hindi phrase to ask the driver	11

Methodology

Method	Think-aloud protocol + task-based testing + post-test interview + SUS questionnaire
Duration	35–40 minutes per session
Setup	Figma prototype on the participant's own phone; screen recording with consent
Metrics	Task completion rate, time on task, error count, SUS score, satisfaction

PHASE 5 · USER FEEDBACK

Participant by participant

Detailed feedback captured during think-aloud sessions on the actual TransitFlow screens.

P1 · Roodren — 20 · UPES · Gujarat · 3 months in Dehradun

6/7 tasks · 36 min · **SUS 78** · struggled on S5, S8

Where they struggled: On Boarding Confirmation (Screen 5) he did not realise the 'Reliability' bar was tappable for more information, and boarded with Medium confidence without checking. On Signal Lost (Screen 8) he pressed WAIT immediately without reading the 'Possible situations' list, missing valuable context.

What they liked: The 98% High Confidence display on Live Journey — 'this number gives me peace of mind, I trust it'. The Critical Mismatch screen impressed him because he had once boarded the wrong Vikram in real life.

> *"The confidence percentage is amazing. I have never seen any transport app show actual numbers like this. It feels honest, not fake."*
>
> **— P1, Roodren**

P2 · Aarav — 19 · Graphic Era · Bihar · 2 months in Dehradun

5/7 tasks · 42 min · **SUS 68** · struggled on S3, S4, S11

Where they struggled: He confused 'Signal Fluctuating' (Screen 3) with 'Low Data' (Screen 4) — 'these mean the same thing to me'. On Feeling Unsure (Screen 11) he did not recognise 'jaoge' as Hindi and assumed it was the name of a bus stop. The pronunciation guide is missing.

What they liked: Offline Mode (Screen 10) reassured him significantly — he noted the same connectivity problem exists in Bihar. He praised the 'Walking Alternative' on Screen 3 as an option most apps would never offer because it isn't monetisable.

> *"Show me how to actually pronounce the Hindi words — 'ISBT jaoge' — I do not know how to say it. Add a speaker button."*
>
> **— P2, Aarav**

P3 · Meera — 21 · DIT · Tamil Nadu · 5 months in Dehradun

7/7 tasks · 29 min · **SUS 85** · struggled on S6

Where they struggled: Her only difficulty was on Critical Mismatch (Screen 6) — she saw the 'BOARD ANYWAY' button and was tempted to tap it just to see what would happen. She felt this destructive option should be more hidden, and that the red 'CRITICAL MISMATCH' label was too text-heavy.

What they liked: As a Tamil speaker who had been picking up Hindi for five months, she found the Hindi phrase card on Screen 11 very useful. She also liked the Confidence Journey 'Stable' indicator on the Arrival screen — 'feels like a fitness app, motivating'.

> *"The Critical Mismatch screen is great but the 'Board Anyway' button should be smaller. Right now it looks equally important as 'Recheck'."*
>
> — P3, Meera

P4 · Karan — 22 · UPES · Punjab · 8 months in Dehradun

6/7 tasks · 34 min · **SUS 72** · struggled on S2, S5

Where they struggled: On Live Journey (Screen 2) the multi-line journey timeline at the bottom confused him — 'I see times but not which is current'. On Boarding Confirmation (Screen 5) he entered route 7 correctly but the screen still said Medium confidence, which left him unsure whether to trust his own input.

What they liked: The eco-savings figure (2.4 kg CO_2) on the Home screen — 'this is good for college students who care about climate'. Quick Favorites (Home, Office, Gym) felt familiar and easy to use.

> *"The medium confidence on boarding even after I entered the right number is strange. If I am sure, why is the system not sure? Confusing."*
>
> — P4, Karan

P5 · Priya — 20 · Graphic Era · Madhya Pradesh · 1 month in Dehradun

4/7 tasks · 48 min · **SUS 64** · struggled on S4, S7, S9, S10

Where they struggled: The most struggle of all participants — and the newest arrival. On Route Verification (Screen 4) she did not understand what 'Live track unavailable' meant. The Live Tracking map (Screen 7) had four different coloured lines that overwhelmed her. On Offline Mode (Screen 10) she thought the app had crashed because of the 'No Internet Connection' headline.

What they liked: Feeling Unsure (Screen 11) was her favourite — 'this looks like a panic button, very clear'. The 'Show this screen to driver' instruction was, in her words, genius — something she would never have thought of on her own.

> *"Please add a beginner mode. I do not know what GPS or signal lost means. Just tell me 'app is searching for the bus, please wait'."*
>
> — P5, Priya

The pattern across participants

Confidence in the product tracked almost exactly with time spent in Dehradun: Meera (5 months) scored 85 and completed every task; Priya (1 month) scored 64 and completed four. The product is currently easiest for the users who need it least — which is precisely the gap the iterations below are designed to close.

PHASE 5 · DATA COLLECTION

Task completion and SUS scores

Task completion matrix

Task	P1	P2	P3	P4	P5	Success
T1 — Plan route + start Companion	Y	Y	Y	Y	Y	**100%**
T2 — Handle Medium signal at stop	Y	Y	Y	Y	N	**80%**
T3 — Verify route 7 before boarding	N	Y	Y	N	Y	**60%**
T4 — Handle Critical Mismatch	Y	Y	Y	Y	N	**80%**
T5 — Recover from Signal Lost	N	N	Y	Y	N	**40%**
T6 — Continue in Offline Mode	Y	Y	Y	Y	N	**80%**
T7 — Use Hindi phrase support	Y	N	Y	Y	Y	**80%**
Total	**5/7**	**5/7**	**7/7**	**5/7**	**3/7**	**74%**

Y = task succeeded · N = required prompting or failed. Note: per-participant totals in the matrix reflect unaided completion; the individual profile cards record P1 and P4 at 6/7 including one prompted success each.

SUS scores per participant

Participant	Score	Rating	Notes
P1 Roodren	78	GOOD	Confident user; missed the reliability bar
P2 Aarav	68	ABOVE AVERAGE	Hindi script unfamiliar; needs audio
P3 Meera	85	EXCELLENT	Highest score, fastest completion
P4 Karan	72	GOOD	Confused by mismatched confidence labels
P5 Priya	64	OK	Newest user; needed a beginner mode
Average	**74**	**Good**	**Above the industry average of 68**

74
average SUS score
above industry avg (68)

74%
task completion rate
26 of 35 tasks succeeded

37.8
average minutes per session
range 29–48 min

9
issues identified
3 critical · 3 high · 3 lower

PHASE 5 · ANALYSIS

Affinity mapping and prioritisation

Issue themes

Theme	Affected screens	Frequency
Confidence labels are inconsistent or unexplained	S2, S3, S4, S5	**5/5 users**
Hindi phrases need audio pronunciation	S11	3/5 users
Destructive buttons are too prominent	S6 — 'Board Anyway'	3/5 users
Technical jargon confuses new users	S8, S10	2/5 users
Live tracking map has too many coloured lines	S7	2/5 users
Reliability bar is not discoverable as interactive	S5	2/5 users

Priority matrix — frequency × severity

#	Issue	Frequency	Severity	Priority
1	Confidence label system inconsistent across screens	5/5	High	CRITICAL
2	Signal Lost flow — users miss the context list	4/5	High	CRITICAL
3	'Board Anyway' on Critical Mismatch is too prominent	3/5	High	CRITICAL
4	Hindi phrases lack audio playback	3/5	Medium	HIGH
5	Reliability bar on the Boarding screen is not interactive	2/5	Medium	HIGH
6	Live Tracking map cluttered with route lines	2/5	Medium	HIGH
7	Offline Mode headline causes panic in new users	1/5	Medium	MEDIUM
8	Walking Alternative buried under the main signal card	2/5	Low	MEDIUM
9	Quick Favorites lack an edit option	1/5	Low	LOW

Highlighted quotes

> *"The confidence percentage is amazing. It feels honest, not fake."*
>
> — P1, Roodren

> *"Show me how to actually pronounce the Hindi words. Add a speaker button."*
>
> — P2, Aarav

> *"'Board Anyway' should be smaller — it looks equally important as 'Recheck'."*
>
> **— P3, Meera**

> *"Medium confidence even after I entered the right number is strange and confusing."*
>
> **— P4, Karan**

> *"Please add a beginner mode. Just tell me 'app is searching, please wait'."*
>
> **— P5, Priya**

PHASE 5 · ITERATIONS

Nine design iterations

Each iteration records the problem, the evidence from testing, the proposed change, and the design rationale — so that every interface decision remains traceable to a user observation.

01 · Standardise confidence labels across all screens

CRITICAL

Affects screens 2, 3, 4, 5, 6, 9

PROBLEM

All five users were confused by inconsistent confidence terminology — 'High Confidence 98%', 'Medium Signal Fluctuating', 'Low Confidence Route', 'Low Data', 'LOW RELIABILITY', 'CRITICAL MISMATCH'. Users could not tell whether these were one scale or several.

EVIDENCE

P2: 'Signal Fluctuating and Low Data — these mean the same thing to me.' P4: 'Medium confidence even after I entered the right number is strange.'

PROPOSED SOLUTION

Adopt a single three-tier system across all screens: HIGH (green, 80–100%), MEDIUM (amber, 50–79%), LOW (red, 0–49%). Always show the percentage. Use identical visual treatment everywhere. Replace 'Critical Mismatch' with 'LOW — wrong route detected' for consistency.

DESIGN RATIONALE

Cognitive consistency builds trust. When the same concept appears with five different labels, users stop trusting any of them. A single scale also makes the percentage meaningful — users can develop an intuition for what 70% feels like.

02 · Signal Lost — force users to read the context

CRITICAL

Screen 8 — Signal Lost

PROBLEM

Four of five users tapped WAIT immediately without reading the 'Possible situations' list. The recovery flow exists, but is being skipped under stress.

EVIDENCE

P1 pressed WAIT within 2 seconds without reading. P5 panicked and could not interpret the screen at all.

PROPOSED SOLUTION

Replace the immediate action buttons with a 5-second forced-read pause that highlights the context first; the action buttons then fade in. Add a plain-language explainer at the top: 'The app cannot see your bus right now. This is normal — choose what to do below.'

DESIGN RATIONALE

Under stress, users default to the largest button. A small delay before the buttons appear ensures the context is at least seen, and the plain-language line reduces panic for non-technical users like P5.

03 · Critical Mismatch — demote 'Board Anyway'

CRITICAL

Screen 6 — Critical Mismatch

PROBLEM

Three of five users said the 'BOARD ANYWAY' button looked equally important as 'RECHECK ROUTE' — a tempting destructive action at the highest-risk moment.

EVIDENCE

P3: 'Board Anyway should be smaller — right now it looks equally important.'

PROPOSED SOLUTION

Make 'RECHECK ROUTE' the only primary CTA — large, dark navy, full width. Reduce 'Board Anyway' to a small text link below: 'I am sure this is the right route — board anyway'. Add an 'Are you sure?' confirmation modal before proceeding.

DESIGN RATIONALE

Destructive actions should require explicit effort. A text link versus a button is a major hierarchy signal — it says 'this is unusual'. The modal adds a final check at the moment of highest risk.

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04 · Add audio pronunciation for Hindi phrases

HIGH

Screen 11 — Feeling Unsure

PROBLEM

Three of five participants — particularly Tamil and Bihari speakers — could not confidently pronounce 'ISBT jaoge'. Some did not realise it was Hindi at all.

EVIDENCE

P2: 'Show me how to actually pronounce the Hindi words. Add a speaker button.'

PROPOSED SOLUTION

Add a prominent speaker icon next to each Hindi phrase, playing a clear recording in a Dehradun accent. Below the phrase, add transliteration in the user's preferred script — Devanagari, or English phonetic ('jah-oh-geh').

DESIGN RATIONALE

For non-Hindi speakers, script alone is useless. Audio plus multi-script transliteration covers speakers from any state. This single feature would have converted P2 from struggle to success.

05 · Make the Reliability bar interactive on Boarding

HIGH

Screen 5 — Boarding Confirmation

PROBLEM

Two of five users did not realise the LOW–HIGH reliability bar could be tapped for more information about why confidence was Medium.

EVIDENCE

P1: 'I just saw the bar but did not know I could tap it.' P4: 'I want to know why it is medium — but the screen does not tell me.'

PROPOSED SOLUTION

Add a small chevron or 'Why?' link next to the reliability bar. Tapping expands an inline explanation: 'Only 2 users have boarded this route in the last 30 minutes. More confirmations build confidence.' Visible expand/collapse interaction.

DESIGN RATIONALE

Showing a score without explaining it creates anxiety. Letting users dig deeper builds trust in the algorithm — they understand the system is being honest about its own limitations.

06 · Simplify the Live Tracking map

HIGH

Screen 7 — Live Tracking

PROBLEM

Two of five users found the map with four coloured lines (red, blue, green, orange) overwhelming and could not identify their actual route.

EVIDENCE

P5: 'Too many lines — I do not know which is mine.'

PROPOSED SOLUTION

Show only the user's active route as a single bold blue line. Other transit lines fade to 30% opacity grey. Add one highlighted vehicle pin with a soft pulse animation, and an optional 'Show all routes' toggle for power users.

DESIGN RATIONALE

Mid-journey is the highest-stress moment. The user needs to see one thing — am I on track? Visual noise from other routes adds cognitive load where none is needed.

07 · Soften the Offline Mode headline

MEDIUM

Screen 10 — Offline Mode

PROBLEM

P5 thought the app had crashed when she saw 'No Internet Connection' as a large headline. New users interpret system messages as errors.

EVIDENCE

P5: 'I thought the app had broken — I almost closed it.'

PROPOSED SOLUTION

Change the headline to 'Working offline — saved data is ready'. Use a friendly tone and a positive checkmark icon instead of the dramatic struck-through wifi symbol. Show available features more prominently than disabled ones.

DESIGN RATIONALE

Offline is a feature, not an error. The current screen frames it as a problem, which causes panic. Reframing it as a continuation shifts the emotional tone from broken to capable.

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08 · Add a Beginner Mode for first-time users

MEDIUM

All screens — global setting

PROBLEM

P5 (1 month in Dehradun) failed four of seven tasks because of unfamiliar technical language: GPS, signal, offline, route ID. Newcomers need plain language.

EVIDENCE

P5: 'I do not know what GPS or signal lost means. Just tell me "app is searching, please wait".'

PROPOSED SOLUTION

Add a Beginner Mode toggle in onboarding. When active: 'Signal Lost' becomes 'Bus is hiding — searching for it'; 'GPS' becomes 'Location'; 'Route ID' becomes 'Bus number'; 'Confidence' becomes 'How sure are we?'. Beginner Mode auto-disables after five successful trips.

DESIGN RATIONALE

Plain language unlocks the app for the most vulnerable users — first-time newcomers like P5. Auto-disabling after a learning period prevents the language from feeling patronising once users are experienced.

09 · Reorder the Stop signal — surface the Walking Alternative

MEDIUM

Screen 3 — Clock Tower Stop

PROBLEM

Two of five users missed the Walking Alternative card because it sat below the dominant 'What should you do?' panel and the 'Wait Here' button. Walking is a free alternative and should be discoverable.

EVIDENCE

Both P1 and P4 were observed tapping WAIT without scrolling down to see the walking option.

PROPOSED SOLUTION

Move the Walking Alternative card directly below the confidence indicator, before the action buttons. Use an equal-weight card design and add an estimated walking arrival time (e.g. '8 min · arrives the same time as the bus').

DESIGN RATIONALE

When the system signal is Medium, walking is genuinely competitive. Surfacing it visually equalises the choice — users can pick based on what they want, not on what the screen biases them toward.

Phase 5 completeness check

Testing & conduct

- Six specific testing objectives documented
- Five participants, all out-of-state newcomers
- Sessions on participants' own phones via Figma prototype
- Screen recording obtained with consent; observation notes per participant
- Think-aloud protocol followed throughout; post-test interview conducted
- SUS questionnaire completed by every participant

Data, analysis & iteration

- Completion matrix built — 7 tasks × 5 participants = 35 data points
- SUS scores calculated per participant and averaged
- Time on task recorded per session (range 29–48 min)
- Affinity mapping across all observations — six thematic clusters
- Priority matrix combining frequency × severity — nine issues categorised
- Nine iterations documented (target was three), each with problem → evidence → solution → rationale

PHASE 5 · CONCLUSION

What works, what has to change

What this prototype does well

- Comprehensive uncertainty handling — Signal Lost, Route Deviation, Offline and Critical Mismatch screens are unique and valuable
- Honest confidence percentages (98%, Medium, Low) instead of fake certainty
- Cultural awareness — Hindi phrase support and the 'show this to the driver' instruction
- Emergency support — the 112 SOS button on the Feeling Unsure screen
- Multi-state journey tracking with visible progress

What needs to change before launch

- The confidence label system must be unified across all 12 screens
- Destructive actions ('Board Anyway') must be visually demoted
- Hindi phrases need audio pronunciation for non-Hindi speakers
- A Beginner Mode is required for first-month newcomers
- Live Tracking must show only the user's route, not all transit lines

Round 2 testing recommendations

After applying the nine iterations, Round 2 should run with five fresh participants — different from Round 1, to avoid familiarity bias — and should validate:

→ Whether the unified confidence scale is correctly interpreted across all 12 screens

→ Whether the demoted 'Board Anyway' button reduces accidental destructive taps to under 5%

→ Whether audio pronunciation actually helps non-Hindi speakers complete T7

→ Whether Beginner Mode improves the task completion rate for the newest users (target 85%+ for P5-equivalent participants)

→ Whether the simplified Live Tracking map is faster to interpret

→ Whether the softened Offline Mode headline reduces panic responses

Final verdict

The 12-screen prototype demonstrates significant maturity in handling real-world data uncertainty — directly addressing the earlier feedback that the flows were too ideal. With an average SUS score of 74 (above the industry average), strong qualitative reception of the confidence indicator and the Critical Mismatch screen, and only nine issues identified across twelve screens, the prototype is solid. The nine proposed iterations will close the remaining gaps — particularly around label consistency, language accessibility, and beginner-friendly framing. Round 2 testing after these changes is recommended before any pilot launch.

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06 · REFLECTION

What I learned across five phases

The reframe was the whole project

Phase 1 produced a competent but generic conclusion: no route info, no timings, no fare transparency. Every Indian city produces that report. The value of Phase 2 was refusing to stop there — re-reading the same nine interviews for *behaviour* rather than complaints, and noticing that every workaround users had invented was a substitute for a missing confirmation signal. Everything downstream, from the colour system to the panic-mode step count, is an expression of that single reframe.

Designing for emotion made the decisions testable, not vaguer

Treating emotional state as a design input could have produced woolly work. Instead it produced sharper constraints: confirm a vehicle in under 60 seconds, deliver a wait-or-switch decision in under 10 minutes, show a fare before boarding, complete a first trip with zero cab switches. Each is measurable, and each traces to a specific person's specific bad day.

Testing exposed the gap I could not see from inside

The most uncomfortable finding was that product confidence tracked with how long a participant had already lived in Dehradun. The app was easiest for the users who needed it least. Priya — one month in the city, four of seven tasks completed — is the actual target user, and the interface was speaking a technical language she had never been taught. Beginner Mode and audio pronunciation exist because of her session.

What I would do differently

→ **Test the vocabulary earlier.** The confidence label inconsistency (five names for one concept) would have surfaced in a 20-minute card sort during Phase 4, long before it cost 5/5 users in Phase 5.

→ **Recruit for the extreme, not the average.** Newest arrivals and lowest digital literacy should be over-represented in the sample, not treated as an outlier at the end.

→ **Prototype the failure states first.** The screens that carried the most value — Signal Lost, Critical Mismatch, Offline — were also the ones designed last and tested least well.

→ **Keep one product name.** The RideMate → TransitFlow shift mid-project cost clarity in the documentation, and would cost more in a real team.

Next steps

Step	What it involves
Apply iterations	Implement the nine documented changes in the Figma prototype, starting with the three critical ones
Round 2 testing	Five fresh newcomer participants, validating the six questions listed in the Phase 5 conclusion
Field validation	Accompany students on real, unfamiliar Vikram routes — measuring cab-switch rate and self-reported confidence, the two metrics that actually define success
Data seeding	Hand-collect route and fare data for the 20 highest-traffic student routes to solve the cold-start problem before any pilot

The project in one sentence: a navigation product for Dehradun's informal transport that measures its own success not in routes displayed, but in whether a newcomer finishes their first journey without needing to ask a stranger, overpay a driver, or give up and book a cab.

Thank you

Closing the Confidence Gap in Public Transport — a five-phase human-centred design project for Dehradun's public transport network.

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